

Analyzing the Consistency Between Customer Ratings and Review Text Across Skincare Product Categories: A Study of Sephora Reviews

This project explores whether the customer ratings to skincare products on Sephora actually match what they write in their reviews. Sometimes a customer gives 5 stars but mentions problems in the review or gives a low rating while writing mostly positive things. The aim is to find these mismatches and see how they differ across skincare subcategories like moisturizers, cleansers, serums, masks, and treatments.

The primary dataset is the "Sephora Products and Skincare Reviews" dataset from Kaggle (<https://www.kaggle.com/datasets/nadyinky/sephora-products-and-skincare-reviews>). It contains over 8000 skincare products and approximately 1 million reviews in CSV format. Each record includes the star rating, review text, product name, brand, price, and user features such as skin type and skin tone. The data is ready to use and does not require any scraping.

To enrich the primary dataset, a second dataset will be collected from Google Trends containing monthly popularity scores for the top skincare brands in the Sephora data (such as LANEIGE, Dr. Jart+, CLINIQUE etc.). This external data will allow the analysis to investigate whether brand popularity influences customer satisfaction and review consistency. Additionally, new features will be derived from the review text: sentiment scores using the VADER tool, review length in word count, a mismatch flag marking reviews where sentiment contradicts the star rating, and price segments dividing products into budget, mid-range, and luxury groups. The analysis will have two parts: first, an exploratory comparison of product subcategories using visualizations and hypothesis tests (t-test, ANOVA), and second, a sentiment rating consistency analysis to identify which categories have the most inconsistent reviews.